Exact Finite-Level Mixing Geometry in Accelerated Collatz Dynamics

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Abstract

We study finite-level mixing geometry for the accelerated Collatz map modulo powers of two. The main result is an exact arithmetic progression description of the support of the induced finite-level transition kernels. This yields an exact total variation formula and an exact mixing time.

We distinguish two different finite-level kernels: the exact Haar kernel and the lift-averaged empirical kernel. The exact support geometry is proved rigorously for the Haar kernel, while numerical spectral observations concern the lift-averaged empirical kernels.

This paper does not claim a proof of the Collatz conjecture. The remaining obstruction is the orbitwise lifting problem: population mixing does not automatically imply orbitwise arithmetic decorrelation.

1 Introduction

Let

$$T(n) = \frac{3n+1}{2^{v_2(3n+1)}}$$

denote the accelerated Collatz map on odd integers.

The purpose of this paper is to study exact finite-level geometry modulo powers of two.

A central point is that geometric support mixing and arithmetic decorrelation behave differently.

Important clarification. This paper does not prove the Collatz conjecture, orbitwise ergodicity, or asymptotic spectral rigidity. The results concern exact finite-level geometry modulo powers of two.

2 Exact Haar Kernel

Fix $K \geq 2$. Let Ω_K denote the odd residue classes modulo 2^K.

We define the exact finite-level Haar kernel

$$P_K^{\mathrm{Haar}}.$$

For each residue class $a \in \Omega_K$, a random odd 2-adic lift is sampled according to normalized Haar measure on $\mathbb{Z}_2^\times$, and projected modulo 2^K after one accelerated Collatz step.

3 Exact AP Geometry

Define
$$S_t(a) = \sum_{s \le t} v_2\Big(3F_K^{s-1}(a) + 1\Big).$$

[Exact finite-level geometry] For every $K \ge 2$, every odd residue class $a \in \Omega_K$, and every $t < K$,
$$\mathrm{supp}(P_K^t(a,\cdot))$$
is exactly a dyadic arithmetic progression with:
$$\text{spacing } 2^{K-S_t(a)},$$
and
$$\text{cardinality } 2^{S_t(a)}.$$
Mixing occurs precisely when the progression fills all odd residue classes.

Proof. The support evolution follows inductively from valuation stability on residue classes modulo 2^{v+1}.

Let
$$v = v_2(3c_0 + 1).$$
If
$$c \equiv c_0 \pmod{2^{v+1}},$$
then write
$$c = c_0 + 2^{v+1}m.$$
Then
$$3c + 1 = (3c_0 + 1) + 3 \cdot 2^{v+1}m = 2^v(u + 2 \cdot 3m),$$
where u is odd.

Hence
$$v_2(3c + 1) = v.$$
Thus all points in the arithmetic progression support share the same next-step valuation. □

4 Exact Total Variation Formula

[Exact TV formula] For the exact Haar kernel,
$$\mathrm{TV}(P_K^t(a,\cdot), \pi_K) = \max\Big(0, 1 - 2^{S_t(a)-(K-1)}\Big).$$

[Exact mixing time]
$$T_{\mathrm{mix}}(K) = K - 1.$$

5 Dual Representation

The exact Haar kernel admits a dual inverse-tree representation.

For finite K, the operator is normalized by truncating the valuation tail.

Define

$$w_{K,v} = \frac{2^{-v}}{1-2^{-K}}.$$

Then

$$(P_K f)(b) = \sum_{v=1}^{K} w_{K,v}\, f(T_K^{-v}(b)).$$

The missing mass corresponds to the truncated valuation tail $v > K$.

6 Lift-Averaged Empirical Kernels

We now distinguish a second family of kernels.

[Lift-averaged kernel] For fixed M, define

$$P_{K,M}(a,b) = \frac{1}{M}\sum_{r=1}^{M} \mathbf{1}\left[T(a+2^K u_r) \equiv b \pmod{2^K}\right],$$

where u_r are sampled high-bit lifts.

These empirical kernels inject additional high-bit uncertainty and appear to model orbitwise decorrelation more faithfully than the exact finite-level Haar kernel.

7 Spectral Behavior

The exact AP geometry and exact TV formula apply to

$$P_K^{\mathrm{Haar}}.$$

However, the numerical spectral experiments concern the empirical kernels

$$P_{K,M}.$$

This distinction is essential.

Observation

Numerical experiments are consistent with a power-law scaling

$$\delta_K \approx C K^{-\alpha},$$

with empirical exponent

$$\alpha \approx 0.39.$$

Its theoretical interpretation remains completely open.

Conceptual distinction

The paper identifies a separation between:

1. geometric support mixing,
2. arithmetic decorrelation.

Support mixing is exact and linear:

$$T_{\mathrm{mix}}(K) = K - 1.$$

The observed spectral decorrelation appears substantially faster.

8 Orbitwise Lifting Problem

The remaining obstruction is the orbitwise lifting problem.

Finite-level exact mixing does not automatically imply orbitwise arithmetic decorrelation.

A hypothetical divergent orbit would need to maintain coherent arithmetic correlations across infinitely many 2-adic scales simultaneously.

Understanding whether such infinite-scale coherence is possible remains open.

9 Future Directions

Promising directions include:

- Walsh–Hadamard analysis,
- transfer operators,
- symbolic renewal systems,
- sparse operator spectra,
- arithmetic decorrelation,
- orbitwise lifting mechanisms.